

Torsional Behavior and Fatigue Assessment of H-shaped Arch Bridge Hanger by Wind-induced Vibration

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Abstract—H-shaped hanger in arch bridge due to its high slenderness ratio and low torsional stiffness is prone to wind-induced torsional motion under low wind speed, thus leading to the fatigue problem at the flange of hanger near the hanger-to-beam connection. The available data is relatively limited to evaluate the fatigue behavior under torsional response of the H-shaped hanger. To assess the impact of this behavior on the service life of the H-shaped arch bridge hanger, a finite element analysis and analytical investigation were undertaken based on the practical engineering of Dongping Bridge. The results obtained from the finite element analysis correlated well with analytical methods. The calculation showed that the fatigue life of the hanger was reduced with the increase of torsion angle. It was determined the angle range for the No.1 hanger of Dongping Bridge without fatigue failure. The torsion angle can be used as the design target of vibration control.

Keywords– Arch Bridge; H-shaped Hanger; Vortex-induced Torsional Vibration; Fatigue Lifetime

I. INTRODUCTION

Hangers transferring the load from the beam to the arch rib are the major components of half through or through arch bridge. H-shaped cross-section hangers are widely used due to convenient manufacture and maintenance. The long and open cross-sectional members have the characteristics of high slenderness ratio, low torsional stiffness and low damping [1,2]. The low torsional stiffness combined with low damping makes them susceptible to torsional motion when wind speed and direction produces aeroelastic instability of the H-shaped hanger [3-6]. These motions have produced high-cycle fatigue damage. A number of arch bridges at home and abroad have been reported to experience wind-induced vibration of the H-shaped hangers or the attached gusset plates [7-9]. In overall, structural measures, aerodynamic measures and mechanical damping measures are usually adopted to improve the wind-resistant stability of the hangers [10-13].

Significant effort has been made to study the aerodynamic characteristics of the H-shaped hangers of arch bridge over the past decades. Maher and Wittig [14] tested several models of long H-shaped members of the Commodore Barry Bridge in wind tunnels for aerodynamic coefficients and for vibration response. Tests results show that vortex excitation and self-excited motion are possible unless frequencies are relatively high and greater damping than the usual structural level is provided. Tuned mass dampers were installed as remedy measures. Ruscheweyh [15,16] presented a refined engineering model for the

estimation of vortex-induced vibrations of slender members. Using the correlation length model, they predicted the amplitudes of the vortex-excited bending vibration of several steel stacks and bridge hangers. Nakagawa [17] studied vortex shedding mechanism of H and I sections and verified the vortex shedding mechanism from the I-section prism is essentially same as that from a circular, triangular or square prism. Niu et al. [18] developed a novel on eddy current TMD device, with antifatigue and maintenance-free in order to improve the hangers' wind-resistance stability. The device met the needs of the vibration control of hangers in long-span steel arch bridges in typhoon areas. Xu et al. [19] established the governing differential dynamic equation for torsional vibration of H-shaped hanger based on the Vlasov theory about the restrained torsion of opened cross section thin-walled beams. The research results can be used as a reference for the optimal design of wind-resistant cables. Zhong et al. [20] carried out a fatigue assessment of one short hanger based on dynamic strain monitoring data of an arch bridge and proposed an engineering approach for fatigue performance assessment of full-field hangers. Raissi et al. [21] employed deep neural networks that were extended to encode the incompressible Navier-Stokes equations coupled with the structure's dynamic motion equation in order to predict the lift and drag forces by vortex-induced vibrations of bluff bodies.

In this paper, a finite element analysis and analytical investigation were carried out to characterize the vortex-induced torsional vibration and the effects on the arch bridge hangers and connections. The research topic synthesizes wind-induced phenomena, torsional member behavior, and fatigue life prediction. Firstly, classical analytical expressions for rotation angle, torque, and stresses in members with idealized boundary conditions subject to torsion were undertaken. Secondly, the finite element model of the hanger was established to study the torsion behavior. Finally, the expected fatigue lives of the hanger were predicted. The study can be used as the design target of vibration control.

II. ENGINEERING PRACTICE

Dongping Bridge is located in Foshan, China. The main bridge is a half-through arch bridge with a central span of 300m. The bridge hangers are all steel member of H-shaped section and the length of the longest No.1 hanger is 40.212m, as shown in Fig. 1. The flanges are solid plate, but the web contains perforations. Dimension of the hanger is shown in Fig. 2. The gusset plate connection joins the flange to the

steel tied beam and the main arch box by high strength bolt as shown in Fig. 3.

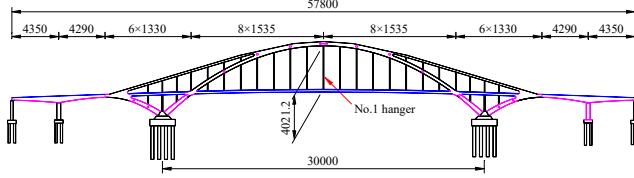


Figure 1. General arrangement of Dongping Bridge (unit: cm)

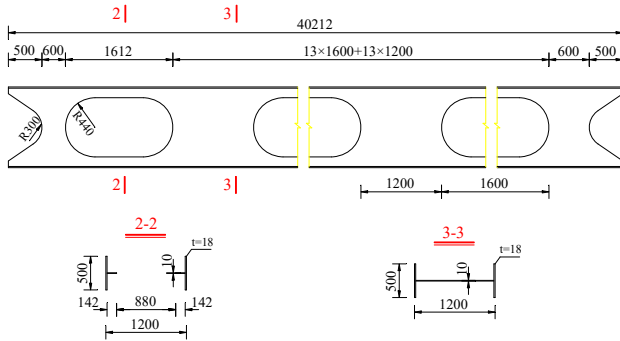


Figure 2. No.1 hanger of Dongping Bridge (unit: mm)

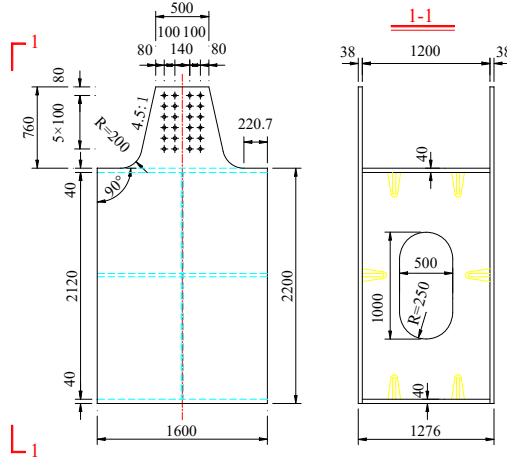


Figure 3. Gusset plate connecting the No.1 hanger to the steel box tied beam (unit: mm)

In 2006, the 13 longest vertical H-shaped hangers in Dongping Bridge showed cracking after a single strong wind event [10]. Damages were discovered on the flanges near the hanger-to-girder connections, as shown in Fig. 4. In order to avoid unanticipated hanger vibrations caused by wind action, several horizontal wind-resistant cables were installed on bridge, as shown in Fig. 5.



Figure 4. Cracked H-shaped hanger in Dongping Bridge



Figure 5. View of the horizontal wind resistant cables on Dongping Bridge

H-shaped arch bridge hangers have still today exhibit wind-induced torsional vibration, included Dongping Bridge. Presently the method and means are not perfect to evaluate the fatigue performance from these torsional responses. It is necessary to predict the fatigue life of existing bridge hangers using theoretical analysis methods and finite element models.

III. THEORETICAL RESEARCH

A. Torsion

To assess the load effects produced by torsional response of H-shaped hanger, the relationship between the rotation and H-section stress must be determined. Partial warping takes place and the twisting moment may be considered to be made up of two parts. One part is a twisting moment, which produces warping on the section in the absence of end restraint. The second part is a twisting moment produced by the lateral shear forces since these forces constitute a couple with moment arm. In the Steel Design Guide Series 9, Torsional Analysis of Structural Steel Members, Seaburg and Carter's approach the torsional problem of different sections with different boundary conditions [22]. Seaburg and Carter's basic equation for the torsional moment resistance of an open cross section is

$$T = GJ\theta' - EC_w\theta'' \quad (1)$$

For convenience let

$$a = \sqrt{\frac{EC_w}{GJ}} \quad (2)$$

so that Equation (1) may be written

$$\frac{T}{EC_w} = \frac{\theta'}{a^2} - \theta'' \quad (3)$$

In Equation (3), θ' is the angel of rotation per unit length, which is shown as the first derivative of the rotation θ with respect to z , where z is the distance measured from the left support along the member as shown in Fig. 6. θ'' is the third derivative of θ with respect to z .

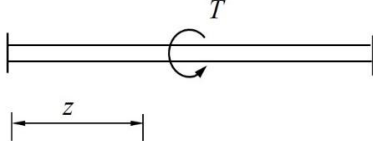


Figure 6. Schematic diagram of a member applied a concentrated torsional moment

The general equation for the rotation θ for a constant concentrated torsional moment T is expressed as

$$\theta = A + B \operatorname{ch} \frac{z}{a} + C \operatorname{sh} \frac{z}{a} + \frac{Tz}{GJ} \quad (4)$$

where A , B and C are constants which are determined according to the boundary conditions. The equation for rotation is presented for different boundary conditions and graphs corresponding to these results can be found in the appendix of [22].

The values for normal stresses due to warping at any point of the flange for an H-section are given as

$$\sigma_{ws} = E W_{ns} \theta'' \quad (5)$$

where W_{ns} is the normalized warping function located at the same point s along the H-section flange, and θ'' is the second derivative of θ with respect to z .

B. Vortex-induced vibration

Vortex shedding can occur when wind flows around a bluff body which disturbs the uniform flow of the wind. The vortex shedding frequency can be determined from the Strouhal relation given as

$$S_t = \frac{f_s D}{U} \quad (6)$$

where S_t is the Strouhal number, f_s is the vortex shedding frequency, D is the dimension of the section perpendicular to the flow and U is the mean flow velocity. The Strouhal numbers for H-sections varied almost linearly from 0.14 to 0.1 as depth/width changed from 0 to 2 [6].

IV. ANALYSIS METHODS OF H-SHAPED HANGER

A. Modeling of the specimens

The dimensions for the model were the same as the half of No.1 hanger of Dongping Bridge considering structural symmetry. The hanger and connecting components were modeled with all relevant cross-sectional properties including perforations as shown in Fig. 7 and the number of divided elements was 35520. To explain the different parts of the FEM, the model has been separated into three regions as shown in Fig. 8.

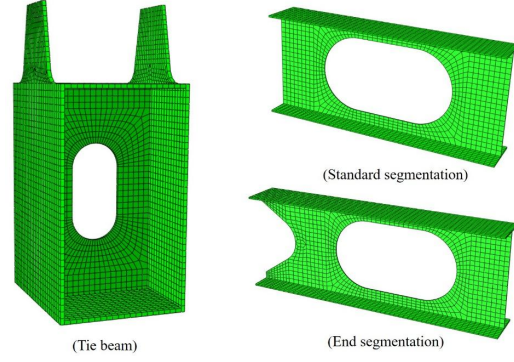


Figure 7. Element division of the model

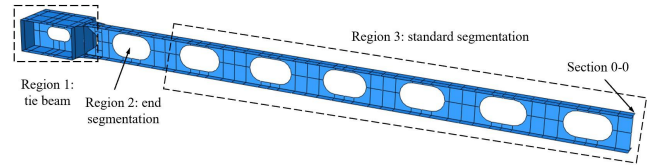


Figure 8. Finite element model of the No.1 hanger with three regions labeled

B. Material properties and element type

The kinematic bilinear stress-strain relationship with full plastic stress was considered for the hanger and part of the steel box tied beam. Tensile coupon tests were conducted to determine the yield and ultimate tensile strength of the steel used, which were 300 and 345 MPa, respectively. Fig. 9 shows the stress-strain relationship for the steel materials. The elasticity modulus (E_s), density (γ), and Poisson ratio (ν) for all steel materials were assumed at 206GPa, 7800 kg/m³, and 0.3, respectively.

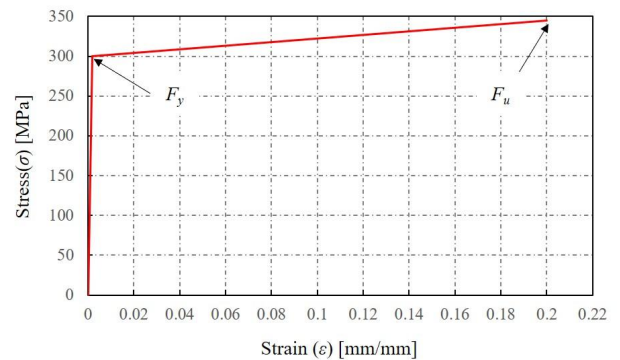


Figure 9. Schematic stress-strain diagram of steel

To mesh the models, the S4R mesh element was used. This is a four-node, doubly curved general-purpose shell having reduced integration with hourglass control and finite membrane strains. Transverse shears are allowed in the elements. For the interaction surfaces between the flange of hanger and the steel box tied beam, a normal behavior was chosen. A linear pressure-overclosure with an arbitrary contact stiffness was selected. This contact stiffness of 8500

N/mm was used for the finite element analyses of No.1 hanger of Dongping Bridge.

C. Element interaction and contact properties

The connections were modeled as face-to-face fasteners, defined by ABAQUS. Rigid multi-point constraint (MPC) fasteners were chosen. The displacements were restrained and rotations were left unrestrained. In Region 1, the connection between the hanger and the tie beam was modeled with three boundary conditions placed as shown in Fig. 10a. The translational degrees of freedom were restrained ($U_x=0$, $U_y=0$, $U_z=0$) and the rotational degrees of freedom were unrestrained. To simulate the strong wall pressing against the reaction box, the entire back side of the tie beam was restrained against movement in the x direction ($U_x=0$). In Region 3, the load was applied to one point near the section 0-0 of the hanger using a special angular displacement as shown in Fig. 16b. The point and the end of the hanger were coupling. The end of the hanger was also restrained against displacements into the x direction ($U_x=0$). The rotational degrees of freedom were unrestrained.

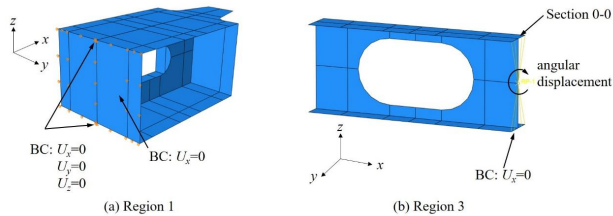


Figure 10. Loading and boundary condition of FEM

D. Results

To study the torsion behavior of the H-shaped hanger, a general static analysis was performed and defined as condition 1. A torque of 95kN·m was applied to the end of the hanger. The predicted torque-angle of twist response at the loaded end was shown in Fig. 11. The final angle of twist was 196°.

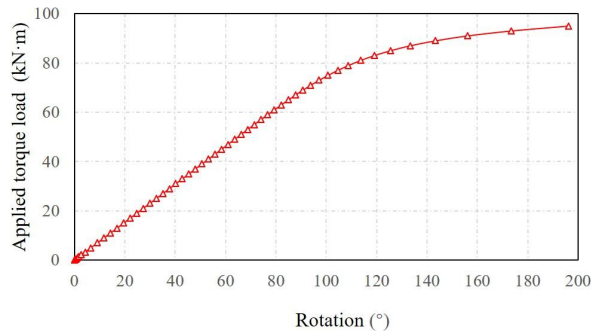


Figure 11. FEM rotation angle-torque plot for the No.1 hanger

When the angle of twist was less than about 60°, the hanger was in the elastic stage. The torsional stiffness of the hanger was calculated to be about 44.129kN·m/rad, as shown in Fig. 12. Then the hanger entered the elastoplastic stage.

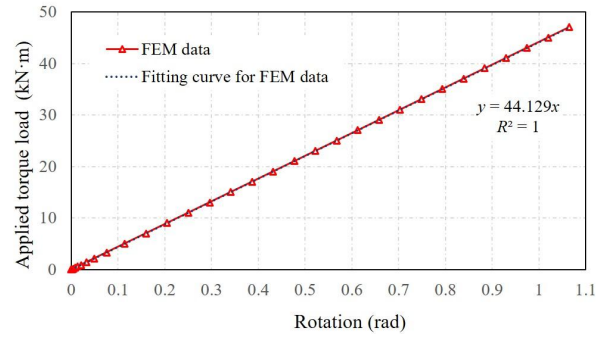


Figure 12. Fitting curve of torsional stiffness by FEM data

First and second yield within the hanger were observed in the top flange of two ends, denoted by A1~A4, as shown in Fig. 13 and 14. The predicted normal stress-twist angle behavior of the element at the first yield location A1 was shown in Fig. 15. First yield occurred at an applied torque of 44.3kN·m and an angle of twist of 57.5°.

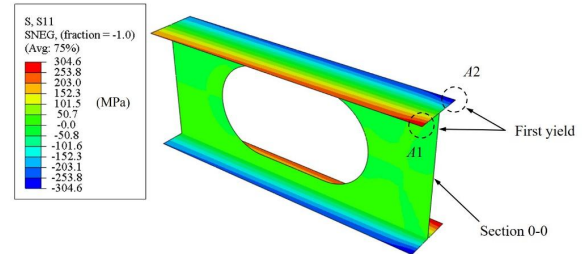


Figure 13. Location of first yield in the FEM

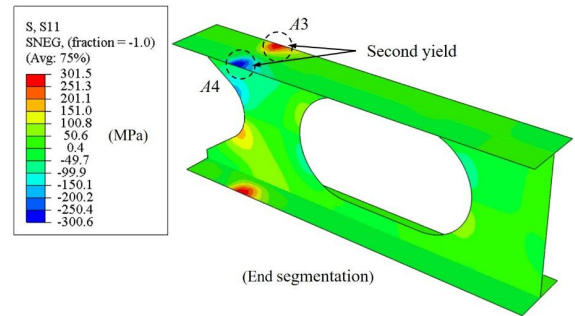


Figure 14. Location of second yield in the FEM

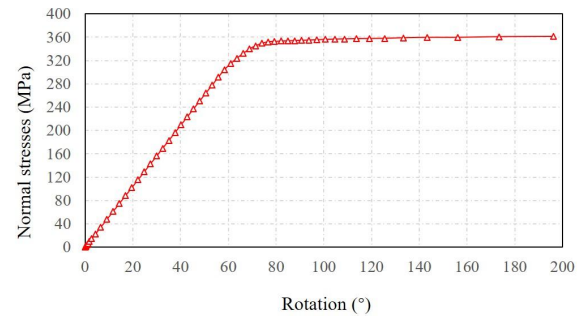


Figure 15. Normal stress- twist angle behavior at A1

To study the deformation and stress distribution of H-shaped hanger in the elastic phase, a general static analysis was performed and defined as condition 2. A forced torsion angle of 10° was applied on the central section 0-0 of the hanger, as shown in Fig. 16. The angle of twist along the span of the H-shaped hanger was calculated using a rearrangement of Equation (4). The angle of twist along the length of the hanger, calculated from the analytical expression and determined with the finite element model, was shown in Fig. 17. The results for the analytical and the finite element angle of twist correspond well.

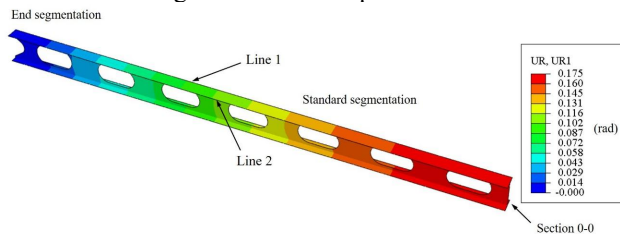


Figure 16. Deformation of the hanger under a forced torsion angle of 10°

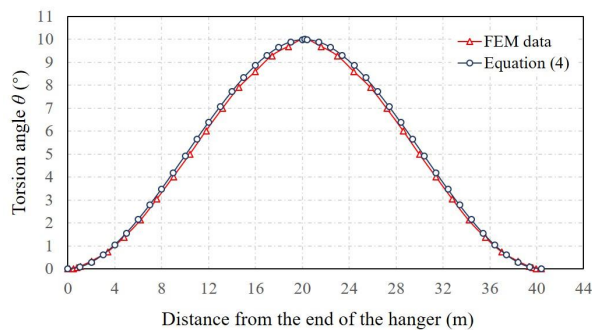


Figure 17. Angle of twist along the length of the hanger

The resulting graph Line1 and Line2 for the top flange were shown and compared to the analytical expression results in Fig. 18 and 19. The maximum normal stress of the hanger was 90.62MPa, which appeared at the end of the flange of the central section of the hanger. Normal stresses in the flange tips were also calculated using Equation (5). The maximum normal stress found in the region for the analytical data was 81.16MPa for the center of the hanger.

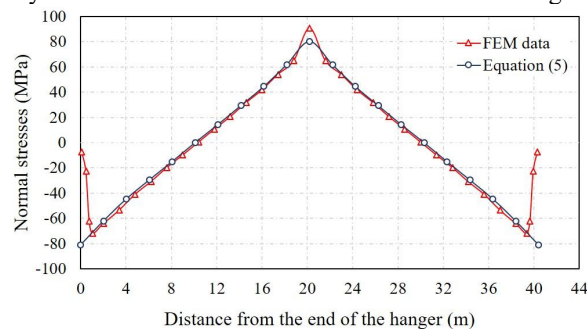


Figure 18. Normal stresses at the Line1 along the length of hanger

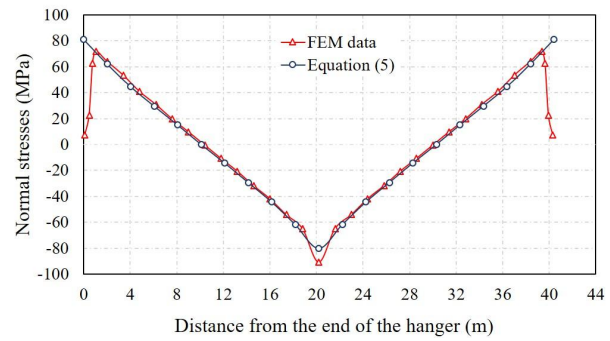


Figure 19. Normal stresses at the Line2 along the length of hanger

V. FATIGUE LIFE ASSESSMENT

In order to study the torsional fatigue of wind-induced vibration of H-shaped hanger, the hanger was checked and calculated according to the relevant content of fatigue design in domestic and foreign steel structure codes. The codes of various countries have given the fatigue stress limits of the corresponding structural details. The relationship between the torsion angle and the fatigue limit. When the wind-induced vibration torsion angle of the hanger does not exceed $\pm 3.68^\circ$, the stress is less than the constant amplitude fatigue limit given in the specification, as shown in Table 1.

Table 1 The relationship between the torsion angle and the fatigue life

Code	Detail Category	Stress range at constant amplitude fatigue limit (MPa)	Fatigue life	The limit of torsion angle ($^\circ$)
BS5400	C	78	107	± 4.35
AASHTO	C	69	2×10^6	± 3.85
Eurocode3	90	66	5×10^6	± 3.68
China	90	66	5×10^6	± 3.68

To predict the total fatigue life of the H-shaped hanger, the torsional natural frequency of the hanger of Dongping Bridge was assumed to be 2.25Hz [10]. The predicted fatigue life, based on different analytically predicted rotational angle ranges for the H-shaped hanger, using Specifications for Design of Highway Steel Bridges, was shown in Table 2. With the increase of the torsion angle, the fatigue life of the hanger was gradually shortened. When the torsion angle reached $\pm 20^\circ$, the fatigue life is only 12490 times. That is to say, the service life of the No. 1 hanger of Dongping Bridge is only 2 hours.

Table 2 Fatigue life prediction of the No.1 hanger

Torsion angle	Stress range	Life cycle	Fatigue life	Fatigue damage	Service life
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(°)	(MPa)	n (1h)	N	D	(h)
±2	17.92	8132	∞	0	∞
±4	35.84	8132	1561233	0.005	192
±6	53.76	8132	462588	0.018	57
±8	71.68	8132	195154	0.042	24
±10	89.60	8132	99919	0.081	12
±12	107.52	8132	57823	0.141	7
±14	125.44	8132	36414	0.223	4
±16	143.36	8132	24394	0.333	3
±18	161.28	8132	17133	0.475	2
±20	179.20	8132	12490	0.651	2

VI. CONCLUSIONS

Vortex shedding for some long H-shaped hanger produced wind-induced torsional vibration in arch bridge. The vortex-induced vibrations represent a high-cycle loading, which very often induces fatigue damage for bridges. Limited data are available to evaluate the torsional response on the fatigue performance of the hangers and connections. To assess the impact of this behavior on the service life of the H-shaped hanger of bridge, a finite element analysis and analytical investigation was undertaken. The finite element model of the hanger was established by using ABAQUS. Based on the theory of linear fatigue cumulative damage and the S-N curve, fatigue damage was evaluated for the hanger under different working conditions.

The following results are obtained in this study.

1. The finite element model was effective in predicting and evaluating the torsional behavior of H-shaped hanger.

2. When the wind-induced vibration twist angle of the hanger does not exceed $\pm 3.68^\circ$, the stress value of hanger is less than the fatigue limit of constant amplitude given in the specification, and fatigue failure of the hanger will not occur. The value of torsion angle can be used as the design target of vibration control.

3. With the increase of torsion angle, the fatigue life of the hanger is reduced gradually. When the torsion angle reaches $\pm 20^\circ$, the fatigue life is only 12490 seconds (about 2 hours).

REFERENCES

- [1] H.F. Xiang, Y.J. Ge. Wind resistance challenges and fundamental research on long span bridges, *Engineering Sciences*, 2011, vol.13, pp.8-21.
- [2] Y.H. An, D.L. Guan, Y.L. Ding, et al. Fast warning method for rigid hangers in a high-speed railway arch bridge using long-term monitoring data, *Journal of Performance of Constructed Facilities* 2017, vol.31, no.6, pp.04017103.
- [3] T. Sarpkaya. A critical review of the intrinsic nature of vortex-induced vibrations, *Journal of fluids and structures*, 2004, vol.19, no.4, pp.389-447.
- [4] S. Cai, W. Chen, M.M. Kashani, et al. Fatigue life assessment of large scale T-jointed steel truss bridge components, *Journal of Constructional Steel Research*, 2017, vol.133, pp.499-509.
- [5] S. Yuan, W.Z. Chen, F. Yang, et al. Reinforcement analysis of rigid hangers for existing old arch bridges: A case study of Ling Bridge, *International Journal of Steel Structures*, 2019, vol.19, no.6, pp.1743-1754.
- [6] P. Keller, C. Higgins, S.C. Lovejoy. Evaluation of torsional vibrations in steel truss bridge members, *Journal of Bridge Engineering*, 2015, vol.20, no.6, pp.04014102.
- [7] C.C. Ulstrup. Aerodynamic lessons learned from individual bridge members, *Annals of the New York Academy of Sciences*, 1980, vol.352, pp.265-281.
- [8] R.J. Wardlaw. Wind effects on bridges, *Journal of Wind Engineering and Industrial Aerodynamics*, 1990, vol.33, no.1-2, pp.301-312.
- [9] J.J. Gu, Y.C. Zhao, K.H. Shao. Application of new TMD to suppressing vortex-shedding vibration of hangers of Jiujiang Bridge over Yangtze River, *China Civil Engineering Journal*, 1994, vol.27, pp.3-13.
- [10] Z.Q. Chen, M.G. Liu, X.G. Hua, et al. Flutter, galloping, and vortex-Induced vibrations of H-section hangers, *Journal of bridge engineering*, 2012, vol.17, no.3, pp.500-508.
- [11] F. Lupi, H. J. Niemann, R. Höffer. Aerodynamic damping model in vortex-induced vibrations for wind engineering applications, *Journal of Wind Engineering and Industrial Aerodynamics*, 2018, vol.174, pp.281-295.
- [12] W.L. Chen, D.L. Gao, S.L. Laima, et al. A field investigation on vortex-induced vibrations of stay cables in a cable-stayed bridge, *Applied Sciences*, 2019, vol.9, no.21, pp.4556.
- [13] J. Naranjo-Pérez, J.F. Jiménez-Alonso, I.M. Díaz, et al. Motion-based design of passive damping systems to reduce wind-induced vibrations of stay cables under uncertainty conditions, *Applied Sciences*, 2020, vol.10, no.5, pp.1740.
- [14] F.J. Maher, L.E. Wittig. Aerodynamic response of long H-sections, *Journal of the Structural Division*, 1980, vol.106, no.1, pp.183-198.
- [15] H. Ruscheweyh. Practical experiences with wind-induced vibrations, *Journal of Wind Engineering and Industrial Aerodynamics*, 1990, vol.33, no.1-2, pp.211-218.
- [16] H. Ruscheweyh, M. Hortmanns, C. Schnakenberg. Vortex-excited vibrations and galloping of slender elements, *Journal of wind engineering and industrial aerodynamics*, 1996, vol.65, no.1-3, pp.347-352.
- [17] T. Nakagawa, R. Nakagawa. Vortex shedding mechanism from prisms having H and I sections, *Journal of Wind Engineering and Industrial Aerodynamics*, 1993, vol.49, no.1-3, pp.197-206.
- [18] H.W. Niu, Z.Q. Chen, X. Hua, et al. Mitigation of wind-induced vibrations of bridge hangers using tuned mass dampers with eddy current damping, *Smart Structures and Systems*, 2018, vol.22, no.6, pp.727-741.
- [19] K. Xu, Y. Zhao, Z.H. Wang. Free torsional vibration analysis of H-section hangers attached with horizontal wind-resistant cables, In *Proceedings of the 2016 World Congress on Advances in Civil, Environmental, and Materials Research*, Jeju Island, Korea, 28 August-1 September, 2016.
- [20] W. Zhong, Y.L. Ding, Y.S. Song, et al. Fatigue behavior evaluation of full-field hangers in a rigid tied arch high-speed railway bridge: Case study, *Journal of Bridge Engineering*, 2018, vol.23, no.5, pp.05018003.
- [21] M. Raissi, Z. Wang, M.S. Triantafyllou, et al. Deep learning of vortex-induced vibrations, *Journal of Fluid Mechanics*, 2019, vol.861, pp.119-137.
- [22] P.A. Seaburg, C.J. Carter. Torsional analysis of structural steel members (Design Guide 9), 2nd ed.; American Institute of Steel Construction, Chicago, 2003.